Road Rage Detection and Prevention System

Group: 74

EBU5608 Product Development and Management

Group Members:

Group Leader

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Summary of Responsibilities

Individual Member 1 (Group Leader) Contribution:

is responsible for overseeing the entire project, coordinating team efforts, and ensuring deadlines are met.
schedules regular team meetings, assists team members with any challenges they encounter during the report preparation, helps find resources and design questionnaires, and aids in creating diagrams.
also manages the final editing, formatting of the report, and specifically finalizes the **Conclusion** on page 26, **List of References** on page 27, and **Appendix** on page 28.

Individual Member 2 Contribution:

finalizes the **Concept Generation and Testing** sections on page 18.
also designs surveys and conducts interviews.

Individual Member 3 Contribution:

finalizes the **Opportunity Identification** on page 6, **Coursework Meeting Logbook** on page 35.

Individual Member 4 Contribution:

finalizes the **Introduction** on page 5, part of **Identification of Customer Needs** on page 12.
is responsible for creating surveys and conducting interviews to gather necessary data.

Individual Member 5 Contribution:

finalizes the **Mission Statement** on page 11, part of **Identification of Customer Needs** on page 12.
also searches for relevant literature and checks the correctness of all citations throughout the document to ensure they meet academic standards.

1. Introduction

As traffic increases and people's stress level rises, traffic accidents are becoming more frequent, often due to road rage. These accidents endanger not only the occupants of vehicles but also pedestrians and other road users. To address this issue, we develop a system that detects the driver's mood. By analyzing real-time traffic condition and monitoring how frequently and intensely drivers use their accelerator and brake pedals, and the tone and content of their spoken words, we can comprehensively assess the driver's mood. This system helps prevent drivers from driving angrily and provides timely reminders, thereby reducing the accident rate.

2. Opportunity Identification

2.1 Establish a Charter

Our team focus on human well-being, and we hope to help human beings solve some problems or improve the happiness of human life through our products. Therefore we five students team up and plan to design a new product in the next two months which hopefully will make people live a happier life.

2.2 Generate and Sense Many Opportunities

Through investigation and group discussions, our team successfully find many opportunities to enhance the well-being of people. Here are 10 ideas.

(1) Stress Assist AI Counseling Platform

(2) Daily Diet AI Optimization Service

(3) Tailored Travel Experience Planner

(4) AR Smart Fitting and Fashion Assistant

(5) Road Rage Detection and Prevention System

(6) Real-Time Cafeteria Crowd Management System

(7) AI-Enhanced Personalized Learning Hub

(8) Comprehensive Alzheimer's Care and Monitoring Platform

(9) Emergency Health Monitoring and Alert Bracelet for Alzheimer's

(10) Interactive VR Communication Aid for the Hearing and Speech Impaired

2.3 Screen Opportunities

We evaluated the above 10 ideas through group discussions which emphasize two aspects: whether they can create value for society and if they are worth further investigation. After that, we successfully selected 3 promising opportunities to carry on for the next step.

(1) Stress Assist AI Counseling Platform

(2) Daily Diet AI Optimization Service

(3) Road Rage Detection and Prevention System

2.4 Develop Promising Opportunities

2.4.1 Customer Interviews for each of the Three Ideas

We conducted several interviews with different kinds of people and collected their opinions. By analyzing their opinions, we concluded the pros and cons of the three products.

Idea I: Stress Assist AI Counseling Platform

Advantages:

1. It utilizes the pressure testing feature of the Apple Watch, providing users with a more intuitive way to monitor and manage stress, enabling better understanding of users' psychological states.

2. It offers an online psychological counseling platform, providing convenient, on-demand professional psychological support to users, helping to alleviate stress and anxiety.

Disadvantages:

1. Application of technology in mental health requires careful consideration, as improper design or use may have negative consequences, such as privacy infringement or inappropriate intervention.

2. While providing an online counseling platform, it lacks the face-to-face interaction, which may impact some users' trust and the effectiveness of psychological counseling.

Idea II: Daily Diet AI Optimization Service

Advantages:

1. It offers a comprehensive diet planning service based on professional nutrition knowledge, helping users make informed decisions about their diet to maintain a healthy lifestyle.

2. It utilizes the Apple Watch to track calorie consumption, providing users with real-time feedback on their dietary habits and progress towards their goals.

Disadvantages:

1. The accuracy of calorie tracking and dietary suggestions may vary depending on the quality of the data and algorithms used, potentially leading to inaccurate recommendations.

2. Reliance on technology and artificial intelligence may lead to a lack of human touch in dietary counseling, potentially overlooking individual preferences and dietary restrictions.

Idea III: Road Rage Detection and Prevention System

Advantages:

1. It reduces the incidence of traffic accidents, ensuring the safety of drivers and passengers as well as pedestrians.

2. It offers a more comfortable driving experience by addressing and mitigating emotional disturbances.

Disadvantages:

1. It may increase the cost of building cars for car companies, and the price of vehicles for customers may also rise.

2. The system's need to monitor and analyze physiological data of drivers raises privacy issues.

2.4.2 Estimates of Market Sizes

(1) Stress Assist AI Counseling Platform

It is designed for people who feel stressful in their daily lives.

(2) Daily Diet AI Optimization Service

It is designed for people who want to control their diet and live a healthier life.

(3) Road Rage Detection and Prevention System

It is designed for all drivers, which means a quite large market.

2.5 Select Exceptional Opportunities

In order to find the best opportunity, we estimate the three ideas above by using the R-W-W framework.

2.5.1 Is the opportunity real?

(1) Stress Assist AI Counseling Platform

Answer: Yes, as there is an increasing demand for mental health services that are accessible remotely.

(2) Daily Diet AI Optimization Service

Answer: Yes, with growing awareness about health and nutrition, personalized diet planning is increasingly sought after. Moreover, the pressure on employees from their work has greatly increased in recent times.

(3) Road Rage Detection and Prevention System

Answer: Yes, due to increasing vehicle usage and stress, there is a real need for solutions that enhance road safety and reduce incidents of road rage.

2.5.2 Can you win with this opportunity?

(1) Stress Assist AI Counseling Platform

Answer: No, winning might be difficult due to the high competition and the challenge of establishing trust and credibility over digital platforms without face-to-face interaction.

(2) Daily Diet AI Optimization Service

Answer: No, it might be challenging to win due to the difficulty in differentiating the service from numerous existing apps and programs that also offer personalized recommendations, often at low cost or even free.

(3) Road Rage Detection and Prevention System

Answer: Yes, if the system is technologically advanced, highly accurate, and can be integrated with existing vehicle systems.

2.5.3 Is the opportunity worth it financially?

(1) Stress Assist AI Counseling Platform

Answer: Yes, the market is growing and willingness to pay for remote, convenient mental health services is high.

(2) Daily Diet AI Optimization Service

Answer: No, financial success could be hard to achieve as such services might require continuous innovation to keep the user engaged and willing to pay.

(3) Road Rage Detection and Prevention System

Answer: Yes, considering the potential to partner with automotive manufacturers and insurers, leading to a significant revenue stream.

Finally, we choose the Road Rage Detection and Prevention System as our product.

2.6 Reflect on the Result and Process

The development process of our project reveals valuable insights and culminated in selecting the road rage detection and prevention system as our focus. This decision is supported by systematic analysis and feedback integration, which highlight the product’s relevance and potential for market impact.

Our choice of the road rage detection and prevention system is based on its strong alignment with safety needs and technological innovation potential. The financial outlook appears promising due to the possibility of forming strategic partnerships, and the product's relevance is heightened by increasing concerns over road safety.

3. Mission Statement

Our mission is to enhance road safety and driver well-being by developing advanced in-vehicle systems that monitor physiological states and predict emotional responses. We aim to reduce the incidence of road rage and improve driving conditions by delivering innovative solutions that seamlessly integrate with vehicle technologies, promoting a safer and more comfortable driving experience for all.

Product Description

- An automotive built-in system that monitors the driver's physiological state and predicts the driver's mood to reduce the risk of road rage.

Key Business Goals

- Product introduced in December 2nd, 2026
- 40% gross margin
- 12% share of auto safety market by 2030

Primary Market

- New car manufacturers
- Luxury vehicle segment
- Fleet management companies

Secondary Market

- Aftermarket
- Auto insurance companies
- Government agencies and non-profit organizations

Assumptions

- Reliable sensors
- Efficient data processing algorithms
- Can be seamlessly integrated system
- Standardized interface between the system and the vehicle

Stakeholders

- Purchasers and Users
- Car Manufacturers
- Vehicle Parts Suppliers
- Insurance Companies
- Government Traffic Safety Agency
- Technology Developers
- Maintenance and Service Providers

4. Identification of Customer Needs

After we finish the mission statement for the project, we first conclude our target customers and then use the Ulrich and Eppinger's five-steps process to identify their needs.

4.1 Target Customer

Fleet Management Companies	Luxury Vehicle Segment	New Car Manufacturers
Organizations that manage a large number of vehicles for commercial purposes are highly sensitive to the costs associated with accidents. This system can offer them a way to improve safety and potentially reduce insurance costs.	High-end automobile manufacturers often seek to differentiate their products with advanced technology and superior safety features. Our system could be a selling point for them.	Auto manufacturers looking to enhance their vehicle safety could integrate this system into new vehicles to appeal to safety-conscious consumers.

Auto Insurance Companies

Insurers could be interested in the system as a tool to mitigate risks, reduce claims related to road rage incidents, and offer discounted premiums to vehicles equipped with advanced safety technologies.

Government Agencies and Non-Profit Organizations

Entities focused on traffic safety and public health might be interested in promoting or subsidizing the adoption of our system to reduce traffic accidents and enhance road safety.

Aftermarket Accessories Providers

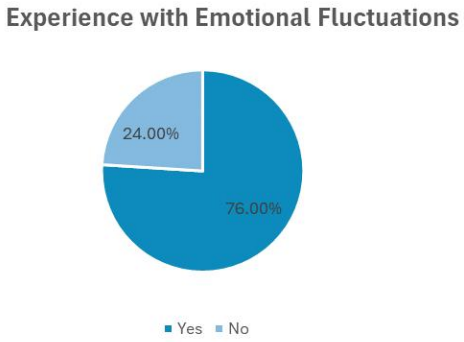
Companies that sell add-ons and upgrades for existing vehicles might see our product as a valuable addition to their offerings, particularly if it can be easily integrated into older vehicles.

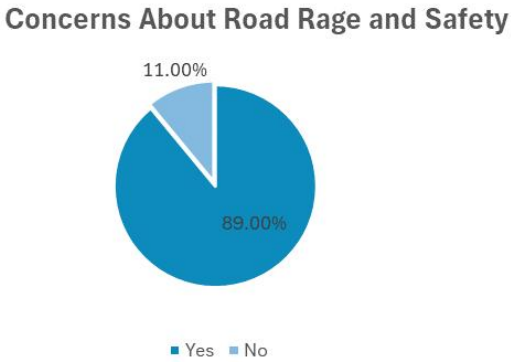
End Users (Drivers and Vehicle Owners)

Ultimately, the main customers are drivers and vehicle owners who are looking for improved safety features, are concerned about road rage, and are interested in technology that can improve their driving experience and safety.

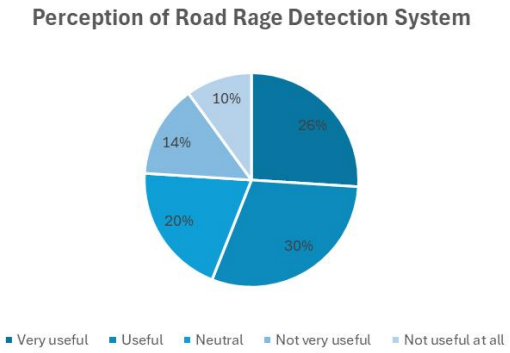
4.2 Gather Raw Data from Customers

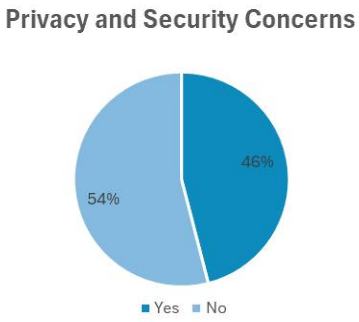
We collected 100 valid data through an online questionnaire (see Appendix A for for full questionnaire). Among them, 76.00% of the respondents mentioned experiencing significant emotional fluctuations while driving or riding due to road conditions, other vehicles, or other factors. Additionally, 89.00% of them fear that road rage or emotional disturbances could lead to traffic accidents.





When it comes to our product, 26.00% of the participants found the system very useful, 30.00% of them thought it as useful and the rest were indifferent or found it not useful. However, 46.00% expressed concerns about privacy invasion.





4.3 Interpret the Raw Data in terms of Customer Needs

Based on surveys and one-to-one interviews based on the youth, middle-aged, and elderly demographics, we have identified the following customer needs (see Appendix B for full interview transcripts).

Question/Prompt	Customer Statement	Interpreted Need
Operation Difficulty	"I might not understand how to operate it."	The control of the system has to be simple to understand and easy to operate.
Privacy Protection	"I'm concerned about my privacy being compromised."	The system ensures thorough protection of user data to guarantee privacy.
Interference Issues	"It may impede my ability to drive safely."	The system can be controlled by the user to enable or disable its usage.
Price	“The car with this system may be expensive.”	The price caters to the consumption level of the masses as much as possible.
Control Methods	"Controlling with a mobile phone might be more convenient."	The system supports multiple methods of control, such as through a mobile phone.
Suggested Improvement	"Would it be better to have different modes to comfort the driver."	The system can be customized by the user to select a mode that suits them.
	"Can I set up the system by myself?"	The system provides users with the option to set personalized modes.

4.4 Organize the Needs into a Hierarchy of Needs

After we interpret the customers' needs, we now use Kano classification which can organize the needs into 4 different types of requirements.

Must-haves

This system indeed features functions that promote calmness in drivers and ensure road safety.

Delighters

The system is straightforward for users to use and does not have any negative impact; The system provides users with the option to set personalized modes; The system ensures thorough protection of user data to guarantee privacy.

Linear Satisfiers

The price of cars with this feature is lower, making users more willing to buy them.

Neutrals

The system supports multiple methods of control, such as through a mobile phone to control switches and modes.

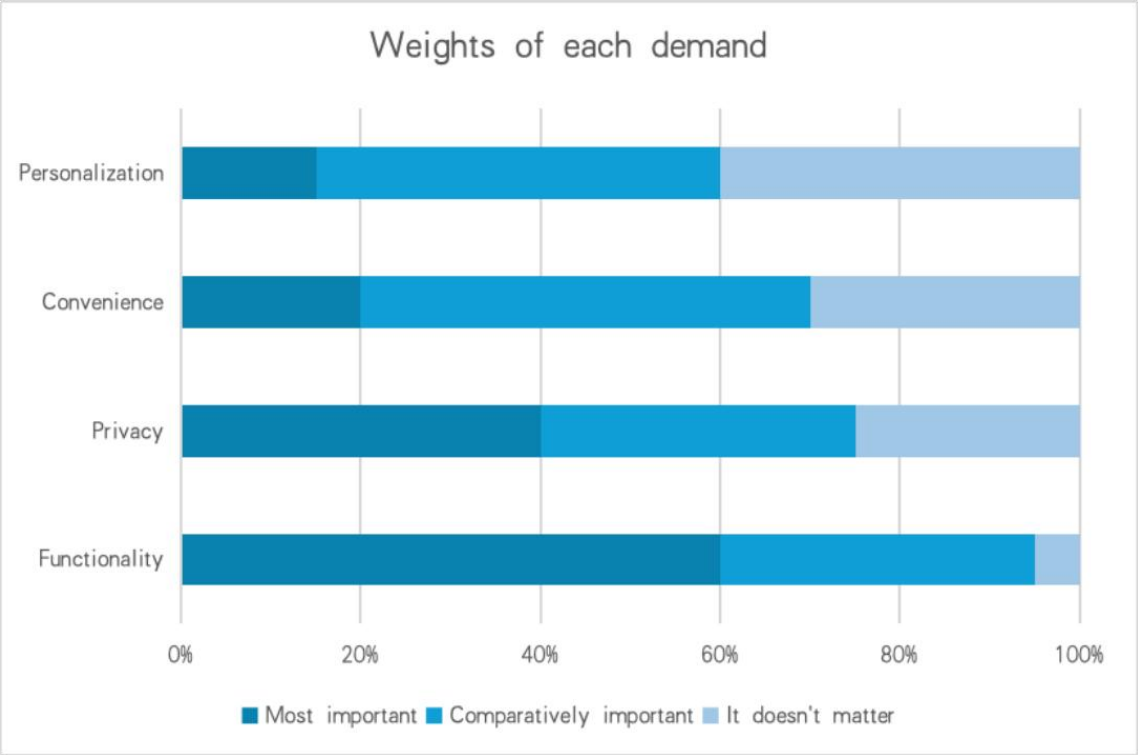
4.5 Establish the Relative Importance of the Needs

Based on our comprehensive survey results and detailed interviews, we have determined the relative importance of the various customer needs associated with our product. It is crucial to prioritize these needs to effectively guide the development process and ensure the product meets the expectation and requirements of our target users.

Our primary focus will be on developing a system that is not only functional and accurate in detecting and intervening in drivers' emotional states but also prioritizes privacy protection.

What's more, it is essential that our product is user-friendly, offering ease of operation through multiple control methods such as mobile devices, ensuring it caters to users' convenience.

Additionally, the ability to personalize the system to suit individual preferences will enhance user engagement, making it more adaptable to various needs.



4.6 Reflect on the Results of the Process

We successfully identified the customer needs and found out what was the most important thing for us to do first. Such prioritization is vital for efficiently allocating resources during development and tailoring our marketing to better communicate the product's benefits.

5. Concept Generation and Testing

5.1 Understanding the Product Idea

In order to develop a technology that can detect a driver's mood, it is first necessary to understand the various emotional states that a driver may experience while driving, and how these emotional states affect driving behavior and safety. This includes, but is not limited to, emotions such as nervousness, anger, fatigue, and excitement.

5.2 Problem Decomposition

We divide the problem into three critical areas:

(1) Road rage detection

(2) Real-time traffic data collection

(3) Reminders and interventions implementation

5.2.1 Road Rage Detection

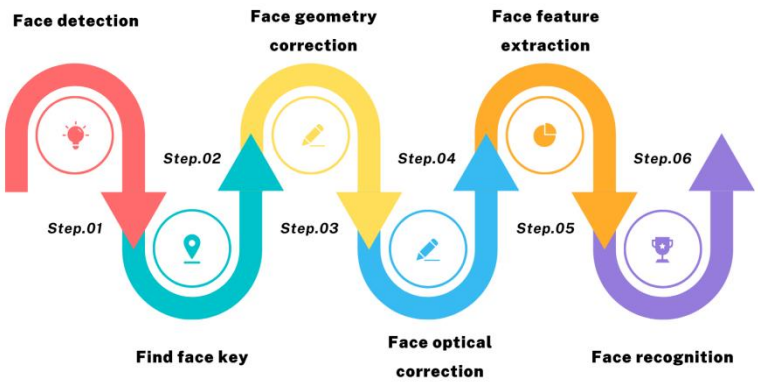
In order to detect road rage or other emotions which may compromise the road safety, we will introduce three kinds of technology, which can work together to ensure an accuracy of up to 81.3% (Liu, 2017).

(1) Face Detection Technology

Driver face detection method combines infrared and depth information. By fusing these two kinds of information, the driver's face area is accurately located. We also employ advanced convolutional networks(ConvNet) and non-maximum suppression(NMS) to enhance face detection accuracy.

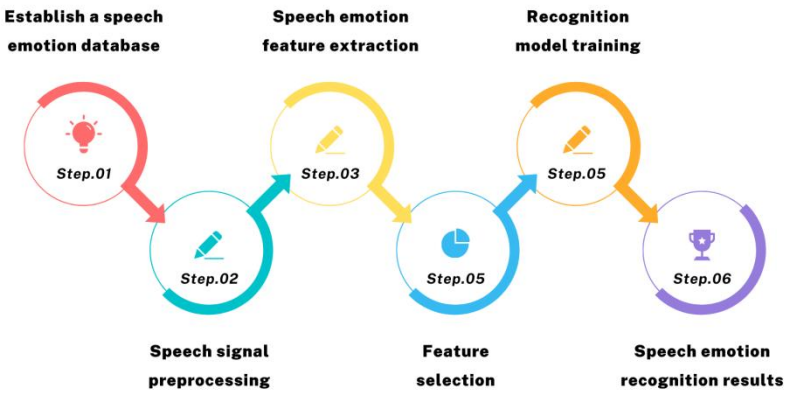
(2) Facial Expression Recognition Technology

We propose a method for recognizing road rage facial expressions which uses a principal component analysis network (PCANet). First, we train the PCANet filter to extract features from the images. Then, we perform hash coding and histogram feature extraction. Finally, we use a support vector machine(SVM) to classify the expressions.



(3) Speech Emotion Recognition Technology

First, we preprocess the speech signal. This involves generating the expression spectrum and performing text vectorization. Next, we extract features from the preprocessed signal by using a convolutional neural network(CNN). Finally, we combine these results to determine whether the driver is experiencing road rage.



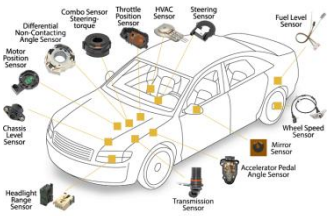
5.2.2 Real-time Traffic Data Collection

In order to detect road information in real time to analyze the specific situation of road rage and decide when to intervene accordingly, we need the help of the following technologies:

(1) GPS and GNSS (Global Navigation Satellite System)

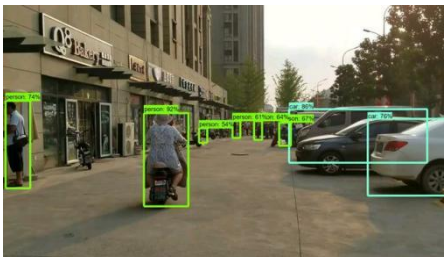
Systems like GPS and GNSS provide accurate position information of the vehicle, including longitude, latitude, altitude, etc. They can update the vehicle's position in real-time, providing basic data for navigation and road condition analysis.

(2) On-board Sensors



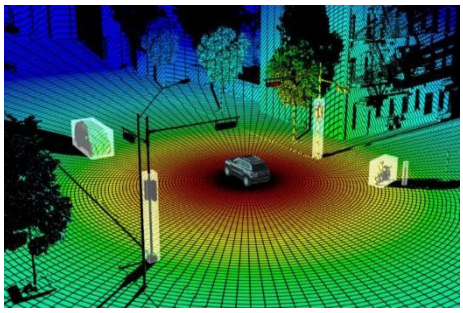
On-board sensors, including speed sensors, accelerometers, and gyroscopes, can monitor dynamic information such as vehicle speed, acceleration, and direction changes in real time. This data helps analyze traffic flow and driving behavior.

(3) Camera and Image Recognition Technology



Vehicle cameras can capture road images, and identify vehicles, pedestrians, traffic signs, and signal lights on the road through image recognition technology, to provide real-time road information, such as traffic congestion, accidents, construction areas, etc.

(4) Radar and LiDAR



Radar can detect the distance and relative speed of objects around the vehicle, while LiDAR generates an accurate three-dimensional map of the surrounding environment by firing laser pulses and measuring the reflected light. These technologies are essential for detecting obstacles around vehicles and assessing traffic conditions.

(5) Vehicle Networking (V2X communication)

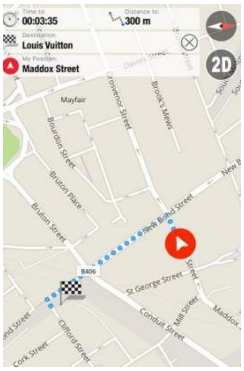
V2X communication technology allows vehicles to communicate with other vehicles (V2V), infrastructure (V2I), pedestrians (V2P), and networks (V2N). Through this communication, vehicles can receive real-time traffic information from other vehicles and infrastructure, such as traffic conditions, accident reports, road condition warnings.

(6) Cloud Computing and Big Data

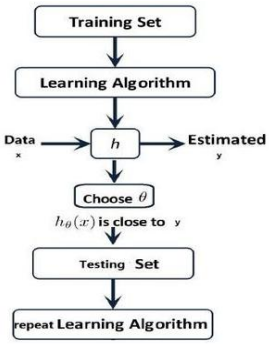
Large amounts of data collected by vehicles can be uploaded to the cloud for storage and analysis. The cloud computing platform can process and analyze this big data to provide more accurate and comprehensive road condition information while supporting complex decision-making and optimization algorithms.

(7) Map and Navigation Software

High-precision mapping and navigation software provide real-time route planning and navigation services. Combined with real-time traffic information, navigation software can dynamically adjust routes to avoid congested areas and provide the best driving path.



(8) Artificial Intelligence and Machine Learning



Through machine learning algorithms, the system can learn from historical and real-time data, predict traffic trends and potential road conditions problems, and provide early warning and advice to drivers in advance.

5.2.3 Reminders and Interventions Implementation

In this section, we will introduce three different methods for implementing reminders and interventions when the drivers' moods are not very good(Wu & Li, 2023).

(1) Visual Intervention

Currently, studies on visual interventions for driver emotions mainly involve environmental lighting, and feedback mechanisms. Environmental light adjusts the driver's mood by altering the cockpit lighting. For example, Jaguar Land Rover's mood detection and soothing system can recognize the nuances of human facial expressions and modify ambient lighting to calming colors.

(2) Auditory Intervention

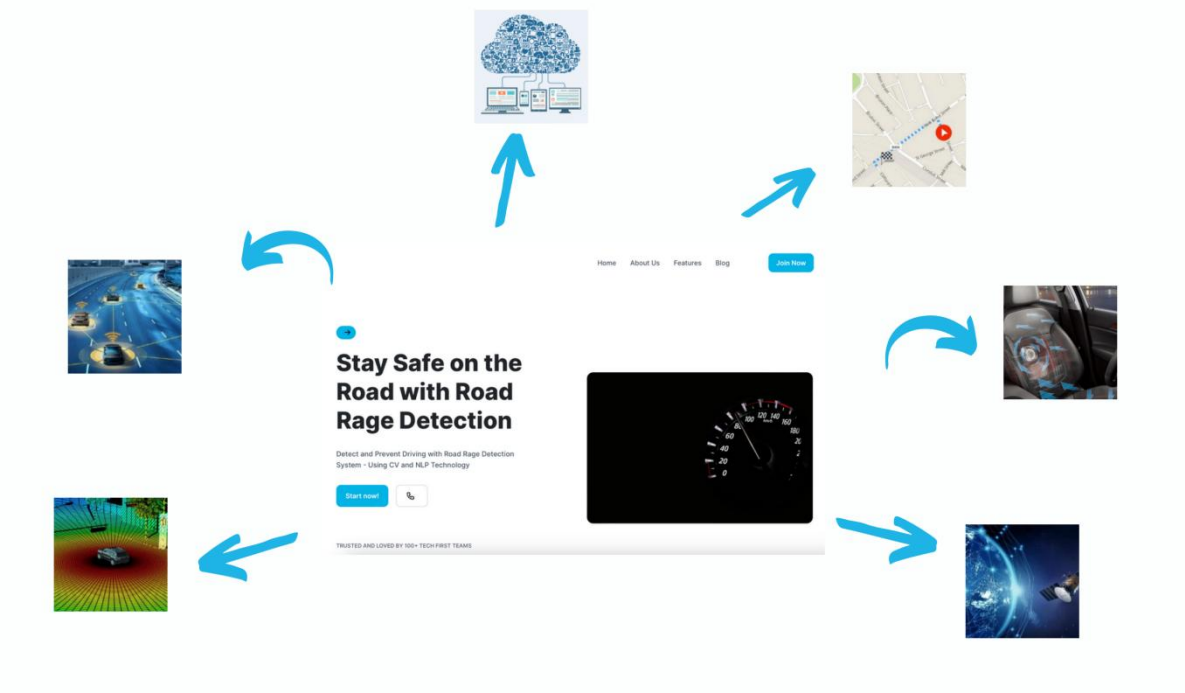
In terms of auditory emotional intervention, the Great Wall WEY2021 is capable of recognizing different facial expressions made by the drivers when entering the cockpit and plays songs or sound prompts that match the emotional context. Similarly, Havel 2021 can actively assess the driver's emotions during the journey and asks whether to close the windows, turn on the air conditioner, or play music, based on the mood detected.

(3) Tactile Intervention

Tactile interventions for managing driver emotions currently include adjustments to air conditioning, steering wheel vibrations, seat massages, or heating. For example, the mood detection and soothing system developed by Jaguar and Land Rover can identify signs of driver sleepiness. If a yawn is detected, the system will lower the air conditioning temperature to help the driver stay alert. Similarly, Havel 2021 will activate seat heating or massage based on the assessed mood of the driver.

5.3 Product Prototype

To visually demonstrate our concept, we have developed preliminary sketches. Below is the interface of our system and some related technologies.



5.4 Concept Testing

5.4.1 Define the Purpose of the Concept Test

The purpose of concept testing is to evaluate the potential market acceptance of the system for drivers, assess its effectiveness in enhancing driver safety and comfort, and identify possible improvements based on user feedback.

5.4.2 Choose a Survey Population

The survey targets a diverse group of participants, including drivers of various ages, and driving experience to ensure comprehensive feedback on the system's usability and effectiveness.

5.4.3 Choose a Survey Format

We utilize online questionnaires to gather detailed feedback. The questionnaire is distributed through automotive forums and other social media platforms (see Appendix C for full questionnaire).

5.4.4 Communicate the Concept

Our Driver Mood Detection System enhances driving safety by accurately identifying a driver's emotional state using advanced sensing technology. Through facial recognition, voice analysis, and environmental sensors, the system detects signs of stress, anger, or fatigue and responds with interventions designed to improve mood and alertness. Interventions include adjusting cabin lighting, playing soothing music, and modifying seat settings for comfort.

5.4.5 Measure Customer Response

The responses from the questionnaire have been evaluated based on the following criteria:

Perceived Usefulness:

The majority of respondents find the mood detection system somewhat to very helpful (65%). This indicates a strong perceived usefulness, particularly with the music and sound prompts and environmental lighting interventions.

Ease of Use:

Most users rate the system as easy to use, correlating with a good to excellent accuracy rating in mood detection by 50% of respondents. This suggests that the system is user-friendly and integrates well into the driving experience.

Potential for Reducing Driver Stress and Road Rage:

A significant portion of respondents (78%) report an improvement in their driving experience, with fewer stress and anger while driving. This outcome points to the system's effectiveness in mitigating negative emotions and contributing to safer driving conditions.

Overall Willingness to Use the System in Daily Life:

With 78% of users noting improved driving experience and 65% finding the system helpful, there is a high willingness among participants to use the system regularly.

5.4.6 Interpret the Results

This section aims to interpret the collected data to estimate overall acceptance and identify factors influencing user acceptance.

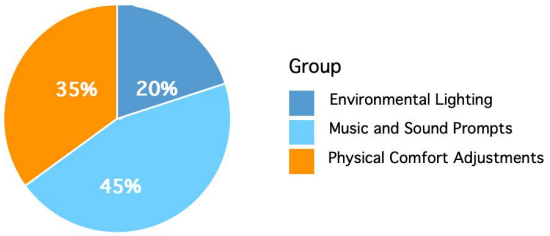
Overall Acceptance:

78% of respondents report a positive impact on their driving experience, highlighting a substantial rate of acceptance. The acceptance is further supported by the 65% of users who find the system helpful in managing their moods.

Factors Influencing Acceptance:

1. Effectiveness of Interventions:

Most people think that Music and Sound Prompts is the most effective way of intervention.



2. Frequency of Negative Emotions:

Users experiencing negative emotions more frequently (often or always, totaling 30%) show higher appreciation for the system, suggesting that the technology is particularly beneficial for drivers who regularly encounter stress or anger while driving.

5.4.7 Reflect on the Results and Process

The overall high acceptance rate and positive impact on driving experience point to a successful reception of the Driver Mood Detection System, indicating a strong market potential. Continual improvements and adaptations based on user feedback will be crucial in refining the system and ensuring broader acceptance and satisfaction among diverse driver demographics.

6. Conclusion

In conclusion, the development process of our road rage detection and prevention system has been insightful and successful. Key points include the importance of integrating user feedback in the design phase and the effectiveness of using the R-W-W framework to evaluate market opportunities. Our product, which predicts and mitigates driver mood disturbances, uniquely addresses the growing concern of road safety through innovative use of technology, showing significant potential to reduce traffic accidents and enhance driver well-being.

Our system stands out by providing real-time mood assessment, integrating seamlessly with vehicle technologies, and prioritizing user privacy and ease of use. This positions it favorably in the automotive safety market, with expectations to achieve a 12% market share by 2030.

Next steps involve extensive market promotion, partnering with car manufacturers and insurers, and continued product refinement based on pilot testing feedback. We will try our best to ensure our system not only meets but also exceeds market and user expectations.

List of References

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Appendix A: Customer Needs Identification Questionnaire

Introduction:

Thank you for participating in this questionnaire about the Road Rage Detection and Prevention System. Your input is essential in helping us understand perceptions and needs related to this system.

Section 1: Demographic Information

1. What is your gender?
   - Female
   - Male

2. What is your age?
   - Under 25
   - 25-45
   - 45-60
   - Over 60

Section 2: Experience and Perceptions

3. Have you experienced significant emotional fluctuations while driving due to road conditions or other factors?
   - Yes
   - No

4. Do you fear that road rage or emotional disturbances could lead to traffic accidents?
   - Yes
   - No

5. Do you find the concept of the Road Rage Detection and Prevention System useful?
   - Very useful
   - Useful
   - Neutral
   - Not very useful
   - Not useful at all

6. Do you have any concerns about privacy invasion with the Road Rage Detection and Prevention System?
   - Yes
   - No

7. Are you attracted to the Road Rage Detection and Prevention System?
   - Yes, very much
   - Yes, somewhat
   - Neutral
   - No, not really
   - No, not at all

Section 3: General Feedback

8. Any additional comments or suggestions for improvement?

Conclusion:

Thank you for your time and valuable input! Your feedback is crucial to our continuous improvement efforts.

Appendix B: Customer Needs Identification Interview Transcripts

Young Participants:

1. Interviewer: Do you think controlling the system with a mobile phone would be more convenient for you?

- Young Participant 1: Absolutely. I'm always on my phone, so it would be much easier.

- Young Participant 2: Controlling it with a mobile phone would fit into my lifestyle better.

2. Interviewer: Do you think having various comfort modes for the driver would improve the experience?

- Young Participant 1: Absolutely, having different modes for comfort would significantly enhance the overall driving experience.

- Young Participant 2: Yes, comfort is crucial, especially during long drives, so having multiple comfort modes would be beneficial.

3. Interviewer: Are you aware that cars equipped with this system may come at a higher price point?

- Young Participant 1: Yes, I've heard they can be expensive, but if it adds value, I'd consider it.

- Young Participant 2: Cost is a factor, but if it's worth it, I'd be willing to invest.

Middle-Aged Participants:

1. Interviewer: Would you prefer if the system had different modes to comfort the driver?

- Middle-Aged Participant 1: Yes, especially since I spend a lot of time commuting.

- Middle-Aged Participant 2: Comfort features are important, especially for longer drives.

2. Interviewer: Would you prefer to have the option to set up the system yourself?

- Middle-Aged Participant 1: Yes, I prefer having the option to set up the system myself. It gives me more control over how I customize it to my preferences.

- Middle-Aged Participant 2: Having the option to set it up myself would be great. It allows me to tailor the system to suit my needs better.

3Interviewer: What are your thoughts regarding the potential impact of this system on driving safety?

- Middle-Aged Participant 1: Safety is my top concern. I would need assurance that the system enhances driving safety instead of compromising it.

- Middle-Aged Participant 2: Safety features are essential for me. I would need to be certain that the system does not compromise my ability to drive safely.

Elderly Participants:

1. Interviewer: Do you have any concerns about using this new technological system? Are you worried it might be difficult to operate?

- Elderly Participant 1: Yes, I'm concerned that I might not understand how to operate it.

- Elderly Participant 2: I'm also worried about it being too complicated for me.

2. Interviewer: How much value do you place on privacy, and are you concerned about potential privacy breaches?

- Elderly Participant 1: Privacy is of the most importance to me. I need to keep my personal information secure.

- Elderly Participant 2: I'm quite cautious when it comes to privacy, particularly with advancements in technology. I want to ensure my data remains protected.

3. Interviewer: Safety is crucial, especially while driving. Do you think this new system might impact your ability to drive safely?

- Elderly Participant 1: I hope it doesn't affect my driving safety negatively.

- Elderly Participant 2: Safety is my top priority. I don't want anything that could distract me while driving.

Appendix C: Concept Testing Questionnaire

Introduction:

Thank you for participating in this important survey! Your feedback is crucial to improving our Driver Mood Detection System, which aims to enhance driver safety and comfort. This questionnaire should take approximately 10 minutes to complete.

Section 1: Demographic Information

1. What is your age?

- Under 18
- 18-24
- 25-34
- 35-44
- 45-54
- 55-64
- 65 and over

2. What is your gender?

- Male
- Female
- Prefer not to say
- Other (please specify): _____

3. How many years of driving experience do you have?

- Less than 1 year
- 1-5 years
- 6-10 years
- 11-20 years
- More than 20 years

Section 2: System Usage Experience

4. How often do you experience negative emotions while driving (e.g., stress, anger, sadness)?

- Never
- Rarely
- Sometimes
- Often
- Always

5. Were you aware of the mood detection system in your vehicle?

- Yes
- No

6. If yes, how would you rate the system's ability to accurately detect your mood?

- Very Poor
- Poor
- Average
- Good
- Excellent

7. How did you feel about the interventions (e.g., environmental lighting, music adjustments) applied by the system based on your mood?

- Very Unhelpful
- Somewhat Unhelpful
- Neutral
- Somewhat Helpful
- Very Helpful

Section 3: Intervention Specific Questions

8. Which intervention did you find most effective?

- Environmental lighting
- Music and sound prompts
- Seat adjustments (massage, heating)
- None

9. Did you find the system interventions improved your driving experience?

- Significantly worsened
- Somewhat worsened
- No impact
- Somewhat improved
- Significantly improved

Section 4: General Feedback

10. Would you recommend this mood detection system to other drivers?

- Definitely not
- Probably not
- Neutral
- Probably yes
- Definitely yes

11. What features would you like to see improved or added to the mood detection system?

12. Any additional comments or suggestions?

Conclusion:

Thank you for your time and valuable input! Your feedback is crucial to our continuous improvement efforts.

Coursework Meeting Logbook

Members of the Team					
Role	**Given Name**	**Family name**	**BUPT number**	**QMUL number**	**Class Number**
Leader					
Project Title	Road Rage Detection and Prevention System				

Date: 22 March 2024
Meeting type: Face-to-face meeting
Topics Discussed: Opportunity identification and scheduling tasks
Tasks to be achieved by next meeting: Develop and distribute questionnaires and begin data collection.
Next meeting (date/time): **27 March 2024**

Date: 27 March 2024
Meeting type: Face-to-face meeting
Topics Discussed: Selection of a product opportunity; formulation of problem and mission statement
Tasks to be achieved by next meeting: Complete the relevant sections of the report.
Next meeting (date/time): 5 April 2024

Date: 5 April 2024
Meeting type: Face-to-face meeting
Topics Discussed: Identification of customer needs and completion of related report sections
Tasks to be achieved by next meeting: Develop a new questionnaire and collect customer perspective data; start drafting the product design.
Next meeting (date/time): 10 April 2024

Date: 10 April 2024
Meeting type: Online meeting
Topics Discussed: Development of product concept and product testing
Tasks to be achieved by next meeting: Finalize the report and start poster design.
Next meeting (date/time): 20 April 2024

Date: 20 April 2024
Meeting type: Face-to-face meeting
Topics Discussed: Finalization of the report and initial poster design
Tasks to be achieved by next meeting: Complete the poster.
Next meeting (date/time): 27 April 2024